

Penetration Testing a Vulnerable Web Application Using OWASP Juice

Shop

Jeytha Sahana Venkatesh Babu

April 7, 2025

Step 1: Install Docker on your Linux machine

```
sudo apt-get update
sudo apt-get install docker.io
sudo docker pull bkimminich/juice-shop
```

Step 2: Pull the OWASP Juice Shop Docker image: `sudo lsof -i :3000`

Output:

COMMAND	PID	USER	FD	TYPE	DEVICE	SIZE/OFF	NODE	NAME
docker-pr	4919	root	4u	IPv4	47694	0t0	TCP	*:3000 (LISTEN)
docker-pr	4924	root	4u	IPv6	47701	0t0	TCP	*:3000 (LISTEN)

Step 3: Get Information about the container

```
sudo docker ps
```

Output:

CONTAINER ID	IMAGE	COMMAND	CREATED
STATUS	PORTS	NAMES	
655d6fb4dd2b	bkimminich/juice-shop	"/nodejs/bin/node /j..."	About a minute ago
Up	About a minute	0.0.0.0:3000->3000/tcp, :::3000->3000/tcp	
reverent_cartwright			

Step 4: Access the OWASP Juice Shop web application by navigating to `http://localhost:3000` in your web browser.

Step 5: Find the correct IP address of the running container: `sudo docker inspect reverent_cartwright | grep "IPAddress"`

Output: "SecondaryIPAddresses": null,
"IPAddress": "172.17.0.2",
"IPAddress": "172.17.0.2"

Step 6: Scan for open ports on the machine running OWASP Juice Shop. Perform the scan without Nmap attempting a ping, use the `-Pn` flag correctly before the target IP. Replace 172.17.0.2 with your own IP.
`nmap -Pn 172.17.0.2`

Output: Starting Nmap 7.80 (<https://nmap.org>) at 2025-04-07 13:49 EDT

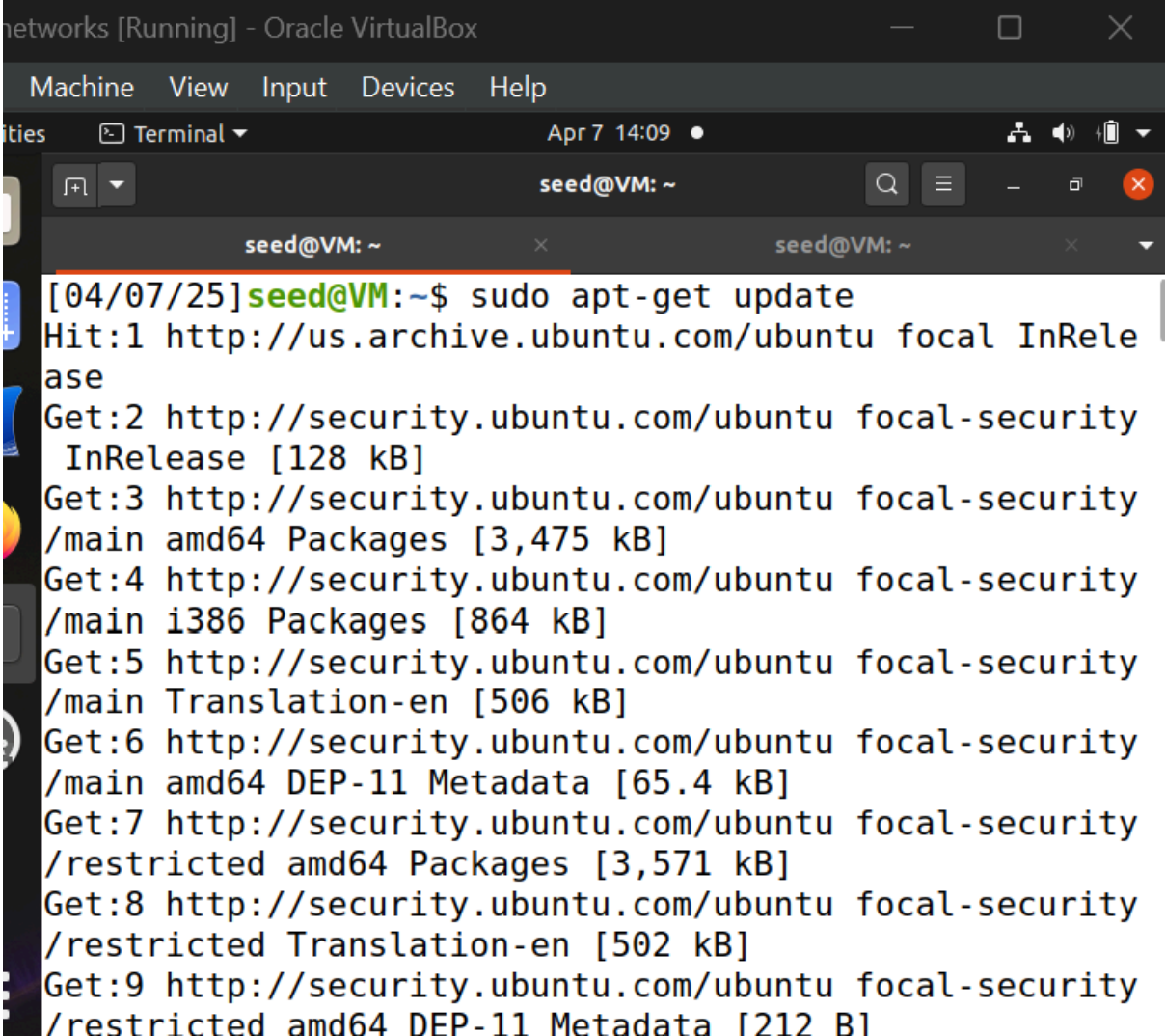
```
Nmap scan report for 172.17.0.2
Host is up (0.00010s latency).
Not shown: 999 closed ports
PORT      STATE SERVICE
3000/tcp  open  ppp
Nmap done: 1 IP address (1 host up) scanned in 0.19 seconds
```

Conclusion: The Nmap scan shows that port 3000 is open on the container, which is the port that OWASP Juice Shop is running on.

Now, you should be able to access the Juice Shop application by navigating to `http://172.17.0.2:3000` in your web browser.

SCREENSHOT

STEP 1:



```
networks [Running] - Oracle VirtualBox
Machine View Input Devices Help
ties Terminal Apr 7 14:09
seed@VM: ~
seed@VM: ~
[04/07/25] seed@VM: ~$ sudo apt-get update
Hit:1 http://us.archive.ubuntu.com/ubuntu focal InRelease
Get:2 http://security.ubuntu.com/ubuntu focal-security InRelease [128 kB]
Get:3 http://security.ubuntu.com/ubuntu focal-security/main amd64 Packages [3,475 kB]
Get:4 http://security.ubuntu.com/ubuntu focal-security/main i386 Packages [864 kB]
Get:5 http://security.ubuntu.com/ubuntu focal-security/main Translation-en [506 kB]
Get:6 http://security.ubuntu.com/ubuntu focal-security/main amd64 DEP-11 Metadata [65.4 kB]
Get:7 http://security.ubuntu.com/ubuntu focal-security/restricted amd64 Packages [3,571 kB]
Get:8 http://security.ubuntu.com/ubuntu focal-security/restricted Translation-en [502 kB]
Get:9 http://security.ubuntu.com/ubuntu focal-security/restricted amd64 DEP-11 Metadata [212 B]
```

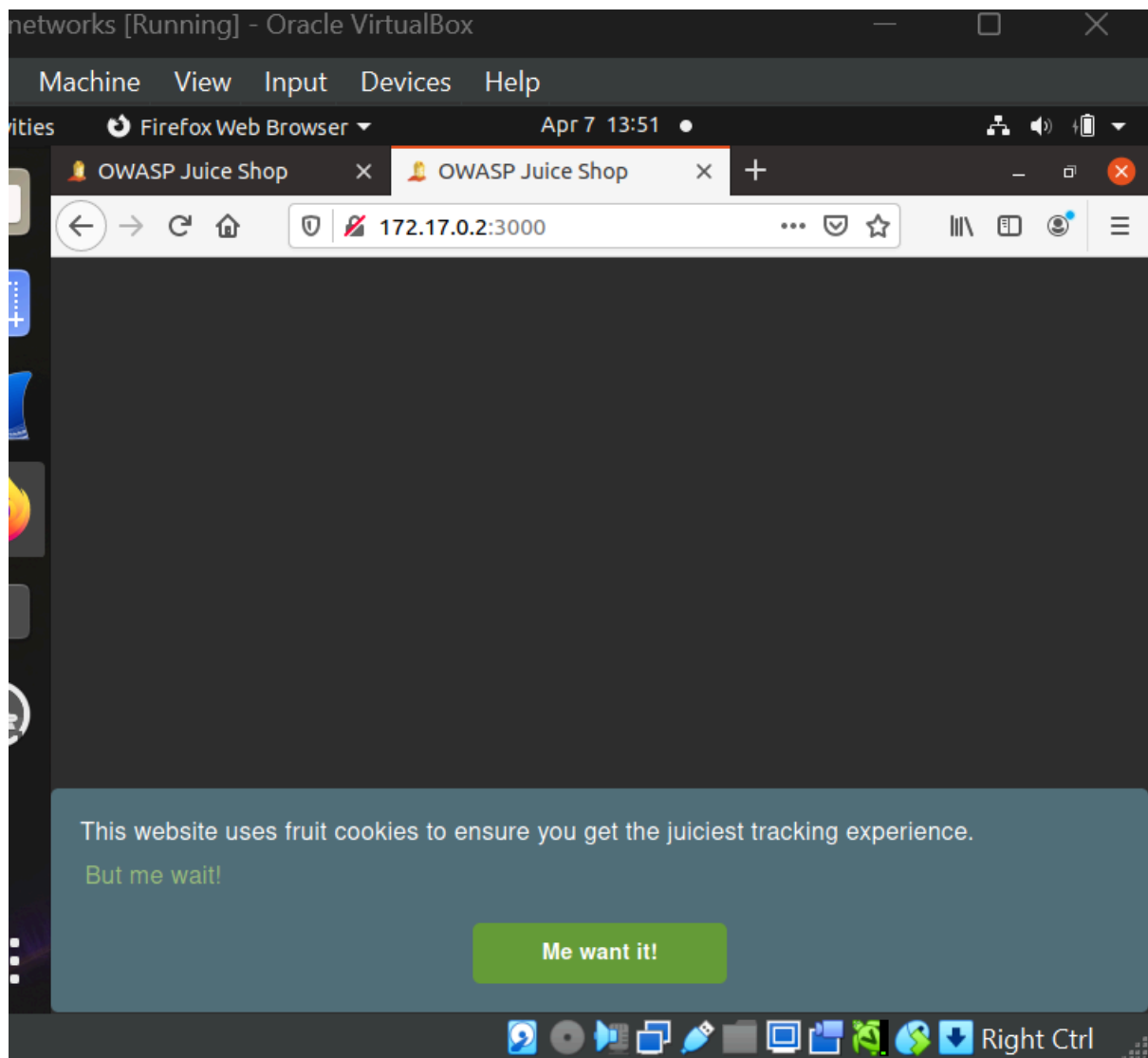
```
networks [Running] - Oracle VirtualBox
Machine View Input Devices Help
vities Terminal Apr 7 14:09
seed@VM: ~
seed@VM: ~ seed@VM: ~
Reading package lists... Done
[04/07/25]seed@VM:~$ sudo apt-get install docker.io
Reading package lists... Done
Building dependency tree
Reading state information... Done
The following package was automatically installed and
is no longer required:
  libfprint-2-tod1
Use 'sudo apt autoremove' to remove it.
Suggested packages:
  aufs-tools btrfs-progs debootstrap docker-doc
  rinse zfs-fuse | zfsutils
The following packages will be upgraded:
  docker.io
1 upgraded, 0 newly installed, 0 to remove and 571 not
upgraded.
Need to get 30.3 MB of archives.
After this operation, 51.7 MB disk space will be freed
.
```

```
networks [Running] - Oracle VirtualBox
Machine View Input Devices Help
vities Terminal Apr 7 14:10
seed@VM: ~
seed@VM: ~ seed@VM: ~
Unpacking docker.io (20.10.21-0ubuntu1~20.04.2) over (
19.03.8-0ubuntu1.20.04.1) ...
Setting up docker.io (20.10.21-0ubuntu1~20.04.2) ...
Processing triggers for man-db (2.9.1-1) ...
[04/07/25]seed@VM:~$ sudo docker pull bkimminich/juice
-shop
Using default tag: latest
latest: Pulling from bkimminich/juice-shop
1c56d6035a42: Pulling fs layer
e33bce57de28: Pulling fs layer
473d8557b1b2: Pull complete
b6824ed73363: Pull complete
7c12895b777b: Pull complete
33e068de2649: Pull complete
5664b15f108b: Pull complete
27be814a09eb: Pull complete
4aa0ea1413d3: Pull complete
9ef7d74bdfdf: Pull complete
9112d77ee5b1: Pull complete
```

STEP 3:

```
networks [Running] - Oracle VirtualBox
Machine View Input Devices Help
Activities Terminal Apr 7 14:11
seed@VM: ~
seed@VM: ~
[04/07/25] seed@VM:~$ sudo lsof -i :3000
COMMAND      PID USER   FD   TYPE DEVICE SIZE/OFF NODE NAME
docker-pr 4919 root    4u   IPv4  47694      0t0  TCP *:3000 (LISTEN)
docker-pr 4924 root    4u   IPv6  47701      0t0  TCP *:3000 (LISTEN)
[04/07/25] seed@VM:~$ sudo docker ps
CONTAINER ID   IMAGE                                COMMAND                  POR
              CREATED                        STATUS                  NAMES
655d6fb4dd2b   bkimminich/juice-shop              "/nodejs/bin/no        0.0
de /j..."    About a minute ago    Up About a minute      .0.0:3000->3000/tcp, :::3000->3000/tcp   reverent_cartwright
[04/07/25] seed@VM:~$
[04/07/25] seed@VM:~$ ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue
state UNKNOWN group default qlen 1000
```

STEP 4:



STEP 5:

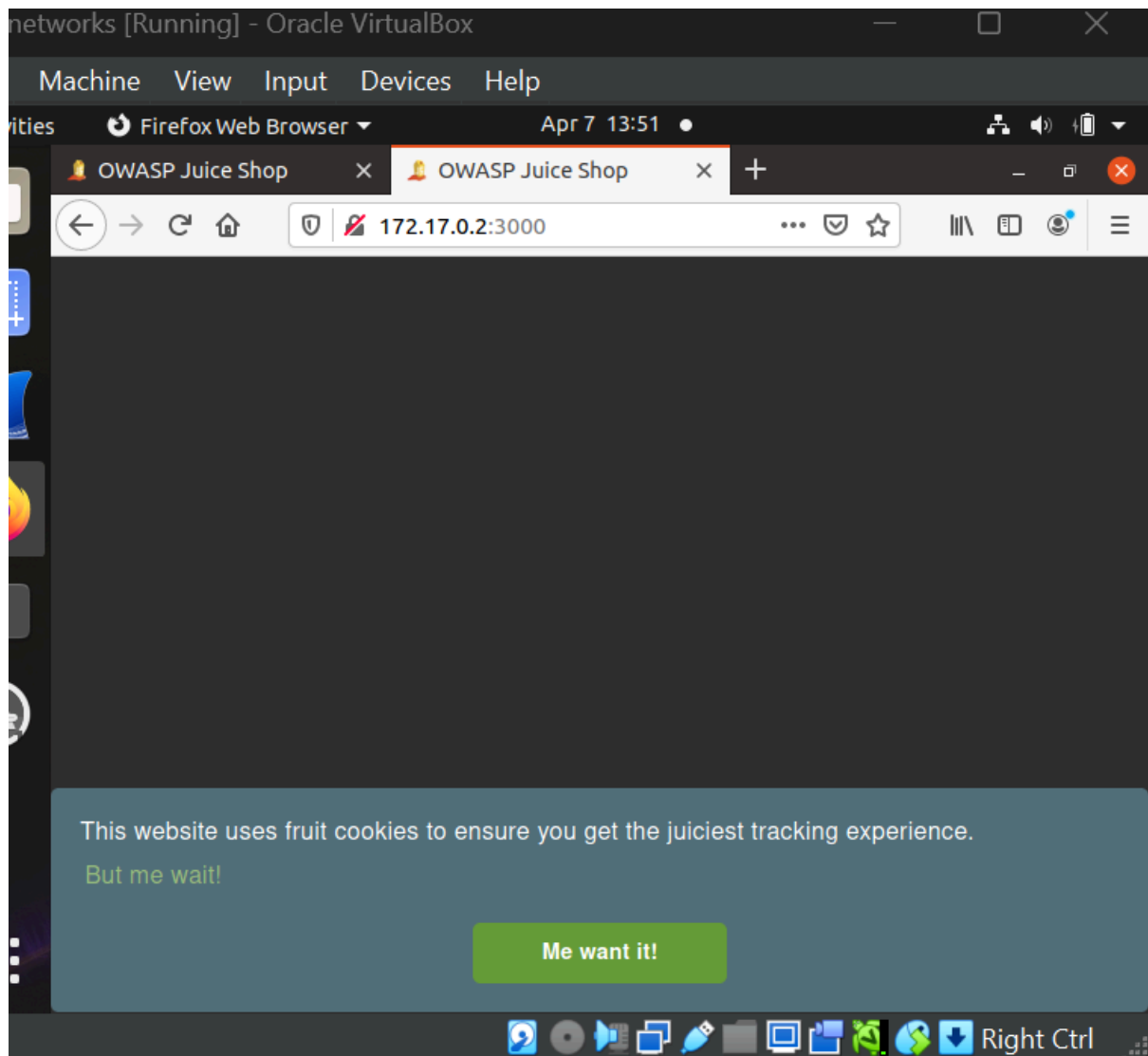
```
[04/07/25]seed@VM:~$ docker ps
CONTAINER ID   IMAGE                                COMMAND
CREATED        STATUS      PORTS
NAMES
655d6fb4dd2b   bkimminich/juice-shop              "/nodejs/bin/no
de /j..."     25 minutes ago Up 25 minutes   0.0.0.0:3000
0->3000/tcp, :::3000->3000/tcp      reverent_cartwright
[04/07/25]seed@VM:~$ sudo docker inspect reverent_cartwright | grep "IPAddress"
      "SecondaryIPAddresses": null,
      "IPAddress": "172.17.0.2",
      "IPAddress": "172.17.0.2",
```

STEP 6:

```
[04/07/25]seed@VM:~$ nmap -Pn 172.17.0.2
Starting Nmap 7.80 ( https://nmap.org ) at 2025-04-07
13:49 EDT
Nmap scan report for 172.17.0.2
Host is up (0.00010s latency).
Not shown: 999 closed ports
PORT      STATE SERVICE
3000/tcp  open  ppp

Nmap done: 1 IP address (1 host up) scanned in 0.19 seconds
```

RESULT:



REFERENCES

0xrajneesh. (2024).

Ethical-Hacking-Projects-for-beginners/Project-2-Penetration-Testing-a-Vulnerable-Web-Application.md at main ·

0xrajneesh/Ethical-Hacking-Projects-for-beginners. GitHub.

<https://github.com/0xrajneesh/Ethical-Hacking-Projects-for-beginners/blob/main/Project-2-Penetration-Testing-a-Vulnerable-Web-Application.md>